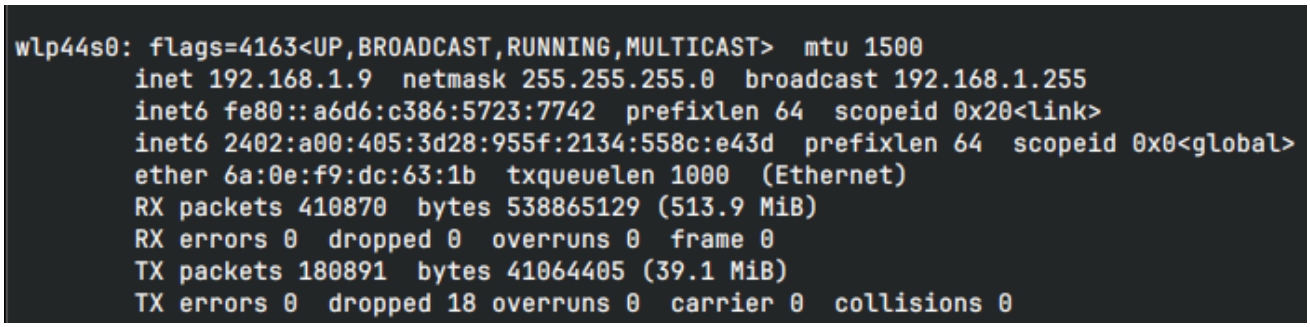
NETWORK TROUBLESHOOTING COMMANDS ASSIGNMENT

1. Open Command Prompt and use ipconfig to find your computer's IPv4 address and default gateway. Take a screenshot of the output.

Command

ipconfig

Network detail	Your result
IPv4 address	192.168.1.9
Default gateway	255.255.255.0



2. Use the ping command to check connectivity between your computer and www.google.com, then note the average response time.

ping www.google.com

Average response time: 14.803/27.835/223.018/28.649 ms

Explanation: Ping sends test packets to the destination and measures how long it takes for replies to return. If requests time out, connectivity may be blocked or unavailable.

3. Run tracert (or traceroute on Linux/Mac) to www.flipkart.com and list the number of hops it takes to reach the destination.

Windows: tracert www.flipkart.com
Linux/macOS: traceroute www.flipkart.com

Number of hops: 9 hops

Explanation: Tracert/traceroute displays the network devices that packets pass through on their way to the destination.

4. Use nslookup to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

nslookup www.spotify.com

IP address returned: 151.101.107.42
Explanation: Nslookup checks DNS resolution by asking a DNS server to translate a domain name into its associated IP address.

5. A friend cannot access Zomato's website from their laptop. Suggest three troubleshooting commands and state what each checks.

Command	What it checks
1. ipconfig	Checks whether the laptop has a valid IP address, subnet information, and default gateway.
2. ping www.zomato.com	Checks basic connectivity and whether the website's host responds to network packets.
3. nslookup www.zomato.com	Checks whether DNS can resolve the Zomato domain name into an IP address.

Troubleshooting sequence: First check the laptop's IP configuration with ipconfig. Next, use ping to test connectivity. Finally, use nslookup to check whether the domain name is resolving correctly. If DNS fails, investigate the configured DNS server; if ping fails, continue checking the local network, gateway, and possible filtering.